



Cisco Catalyst 2960-X

Enterprise Network Infrastructure Recovery & Restoration

Technical Evidence Report | WS-C2960X-24TS-L | Tera Term serial console | Isolated bench

Project scope. Brandon Stevenson manually performed and documented 12 phases of Cisco Catalyst 2960-X secure sanitization, IOS recovery, boot configuration, and post-recovery CLI verification. The screenshots capture his work as each phase was completed.

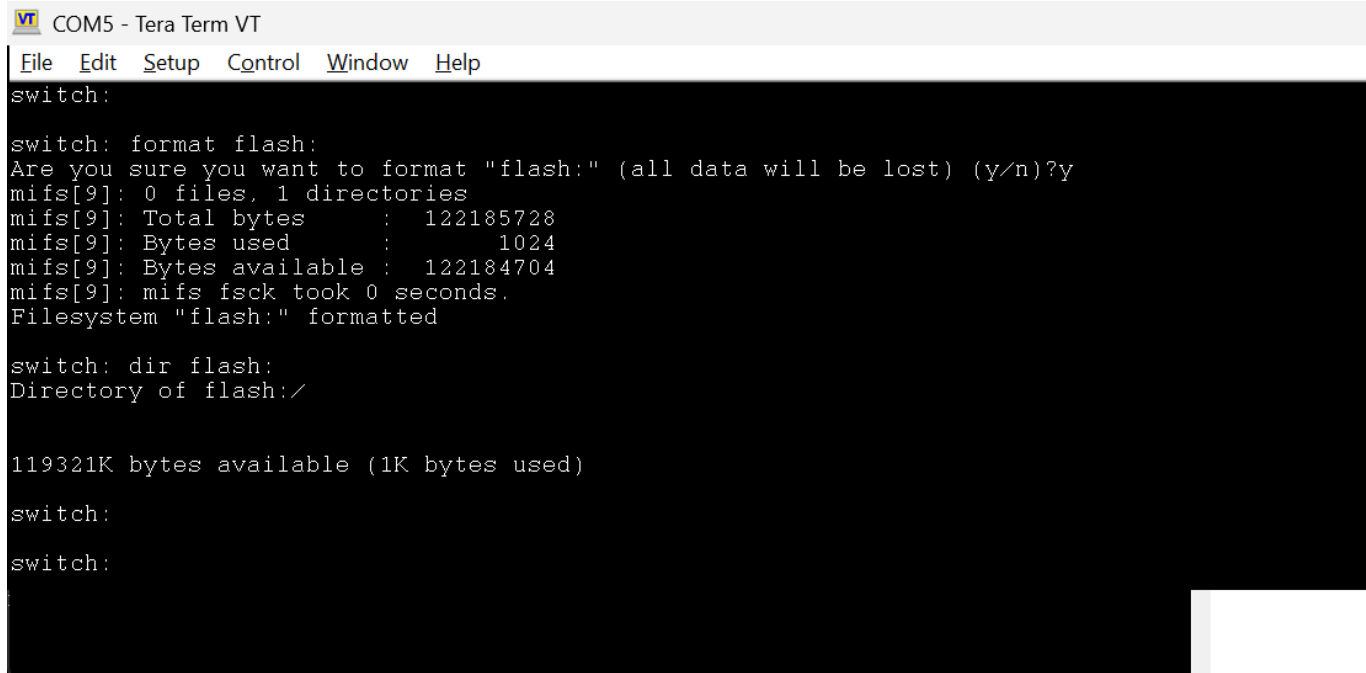
Recorded workflow

01. **Flash formatting and directory check** — format flash::; dir flash:
02. **Bootting approved IOS from USB** — boot usbflash0:c2960x-universalk9-mz.152-7.E11.bin
03. **Secure factory-reset invocation** — factory-reset all secure
04. **FIPS zeroization procedure** — conf t; fips zeroize
05. **Bootloader environment inspection** — set; dir usbflash0:
06. **USB boot after zeroization** — boot usbflash0:c2960x-universalk9-mz.152-7.E11.bin
07. **Flash recovery and persistent boot configuration** — copy usbflash0: flash::; boot system flash:...; copy run start; show boot
08. **Running configuration inspection** — show run
09. **Startup configuration inspection** — show start
10. **VLAN database inspection** — show vlan
11. **VTP and PoE capability check** — show vtp status; show power inline
12. **Device identity validation** — show inv

Evidence walkthrough

Bootloader access and flash preparation

Before Phase 01, I pressed and held the switch's front-panel MODE button during startup to interrupt the normal boot sequence and access the bootloader command prompt (switch:). This provided direct access to flash maintenance commands before Cisco IOS loaded. At the switch: prompt, I ran format flash: to initialize the internal flash file system, then dir flash: to verify its contents.



```
COM5 - Tera Term VT
File Edit Setup Control Window Help
switch:
switch: format flash:
Are you sure you want to format "flash:" (all data will be lost) (y/n)?y
mifs[9]: 0 files, 1 directories
mifs[9]: Total bytes      : 122185728
mifs[9]: Bytes used      : 1024
mifs[9]: Bytes available : 122184704
mifs[9]: mifs fsck took 0 seconds.
Filesystem "flash:" formatted

switch: dir flash:
Directory of flash:/

119321K bytes available (1K bytes used)

switch:
switch:
```

Figure 1. CLI screenshot manually captured by Brandon Stevenson during Phase 01.

02 | Booting approved IOS from USB

Commands observed: boot usbflash0:c2960x-universalk9-mz.152-7.E11.bin

The 'boot usbflash0:...' command tells the switch to load Cisco IOS directly from the attached USB drive. The console then checks the image and starts loading the operating system.

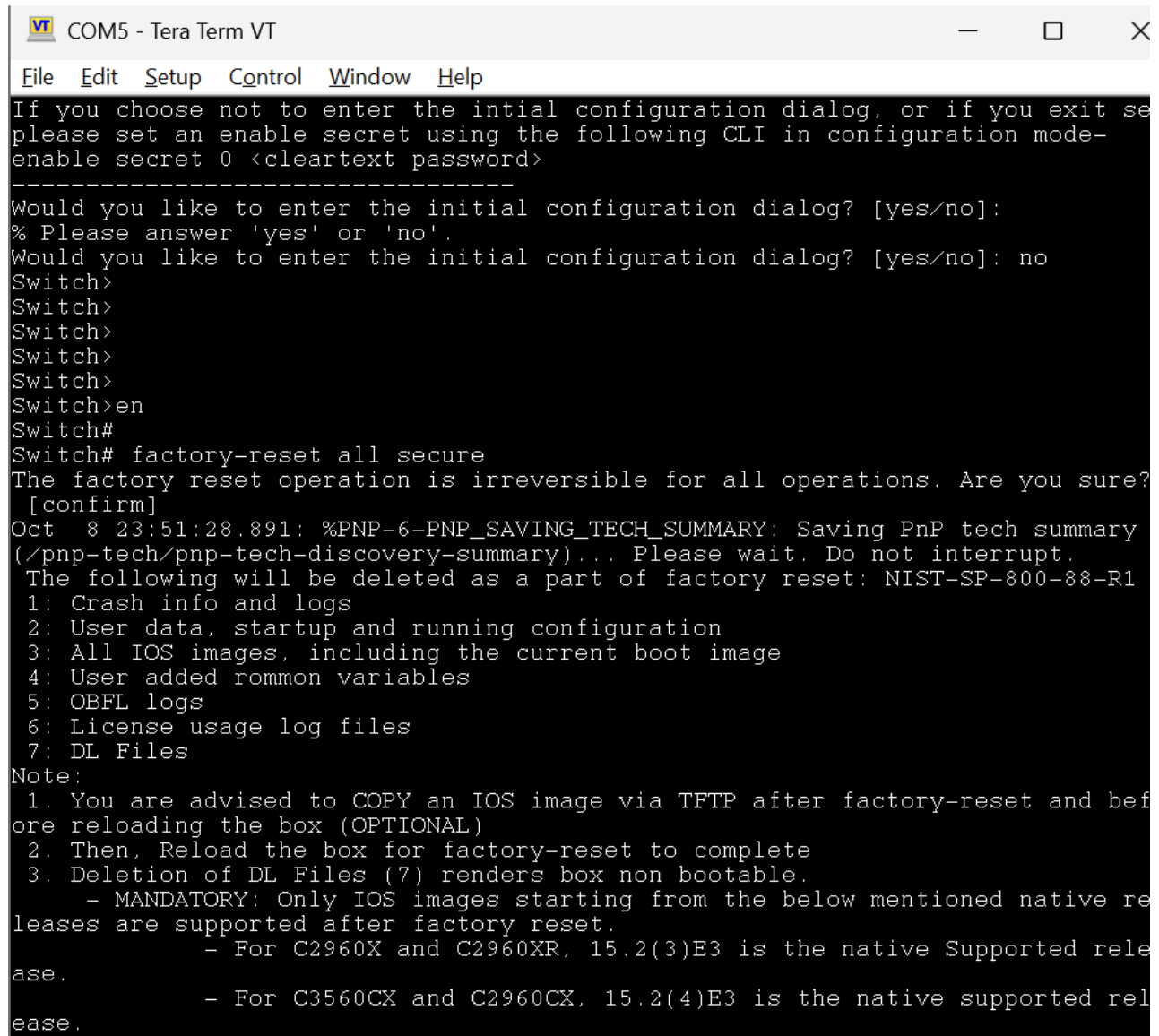
[illegible]

Figure 2. CLI screenshot manually captured by Brandon Stevenson during Phase 02.

03 | Secure factory-reset invocation

Commands observed: `factory-reset all secure`

The 'factory-reset all secure' command starts the switch's secure factory-reset process. The confirmation prompt warns that existing settings and stored data will be erased before the reset proceeds.



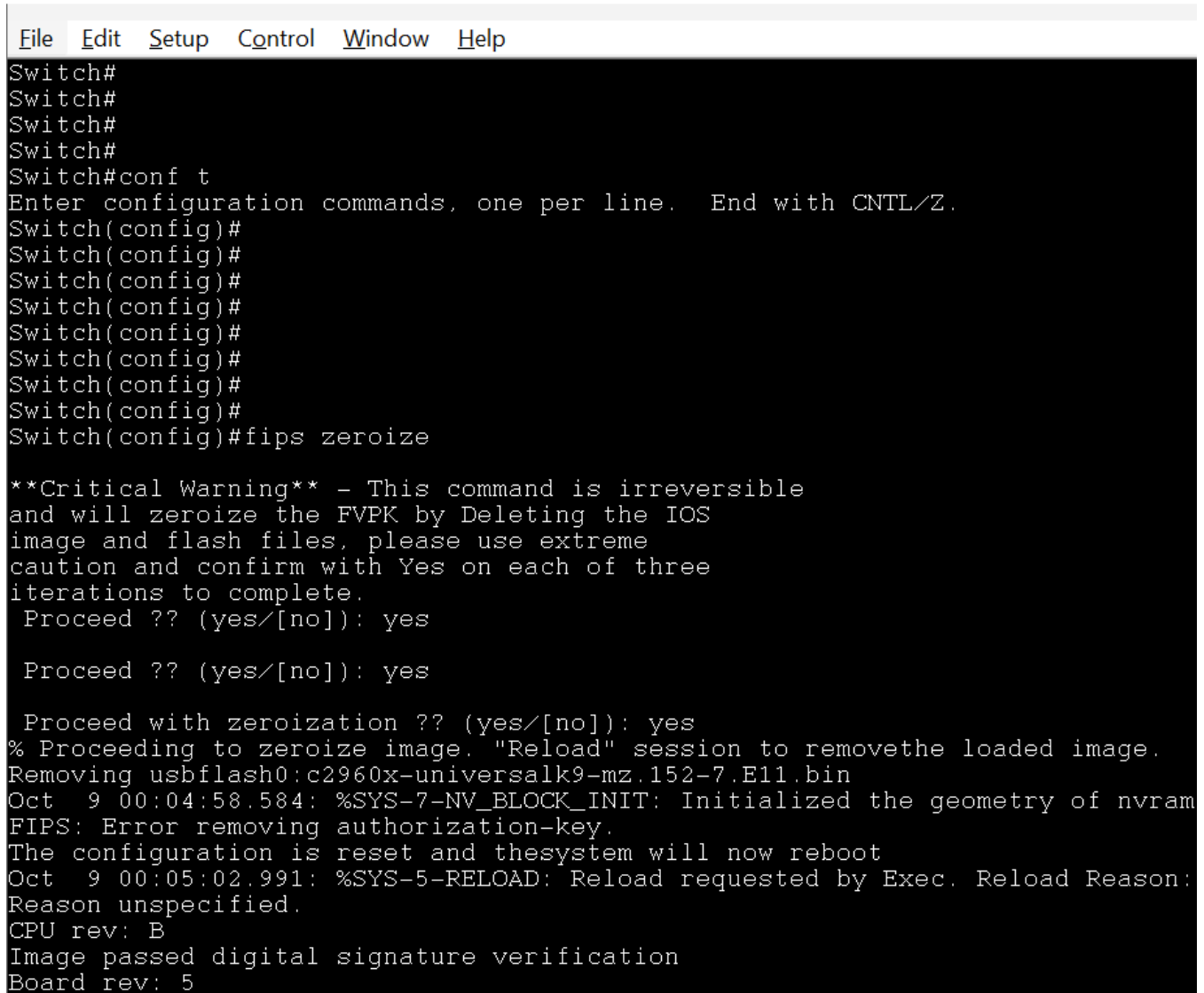
```
COM5 - Tera Term VT
File Edit Setup Control Window Help
If you choose not to enter the initial configuration dialog, or if you exit se
please set an enable secret using the following CLI in configuration mode-
enable secret 0 <cleartext password>
-----
Would you like to enter the initial configuration dialog? [yes/no]:
% Please answer 'yes' or 'no'.
Would you like to enter the initial configuration dialog? [yes/no]: no
Switch>
Switch>
Switch>
Switch>
Switch>
Switch>en
Switch#
Switch# factory-reset all secure
The factory reset operation is irreversible for all operations. Are you sure?
[confirm]
Oct  8 23:51:28.891: %PNP-6-PNP_SAVING_TECH_SUMMARY: Saving PnP tech summary
(/pnp-tech/pnp-tech-discovery-summary)... Please wait. Do not interrupt.
The following will be deleted as a part of factory reset: NIST-SP-800-88-R1
1: Crash info and logs
2: User data, startup and running configuration
3: All IOS images, including the current boot image
4: User added rommon variables
5: OBFL logs
6: License usage log files
7: DL Files
Note:
1. You are advised to COPY an IOS image via TFTP after factory-reset and bef
ore reloading the box (OPTIONAL)
2. Then, Reload the box for factory-reset to complete
3. Deletion of DL Files (7) renders box non bootable.
   - MANDATORY: Only IOS images starting from the below mentioned native re
leases are supported after factory reset.
     - For C2960X and C2960XR, 15.2(3)E3 is the native Supported rele
ase.
     - For C3560CX and C2960CX, 15.2(4)E3 is the native supported rel
ease.
```

Figure 3. CLI screenshot manually captured by Brandon Stevenson during Phase 03.

04 | FIPS zeroization procedure

Commands observed: `conf t; fips zeroize`

Entering 'conf t' opens configuration mode. 'fips zeroize' then initiates the security zeroization process, which clears protected security material; the confirmation warns that this action cannot be undone and may restart the switch.

A screenshot of a network switch's command-line interface (CLI) window. The window has a menu bar at the top with 'File', 'Edit', 'Setup', 'Control', 'Window', and 'Help'. The main area shows a terminal session. The user enters 'conf t' to enter configuration mode. Then, they enter 'fips zeroize'. The system displays a series of prompts and warnings, including a critical warning about the irreversibility of the action and the deletion of the IOS image and flash files. The user responds 'yes' to three prompts. The system then proceeds with the zeroization, showing messages about removing the USB flash image, initializing NV block, and resetting the configuration. Finally, it shows the system will reboot and displays hardware information like CPU rev and board rev.

```
Switch#
Switch#
Switch#
Switch#
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
Switch(config)#
Switch(config)#
Switch(config)#
Switch(config)#
Switch(config)#
Switch(config)#
Switch(config)#
Switch(config)#fips zeroize

**Critical Warning** - This command is irreversible
and will zeroize the FVPK by Deleting the IOS
image and flash files, please use extreme
caution and confirm with Yes on each of three
iterations to complete.
  Proceed ?? (yes/[no]): yes

  Proceed ?? (yes/[no]): yes

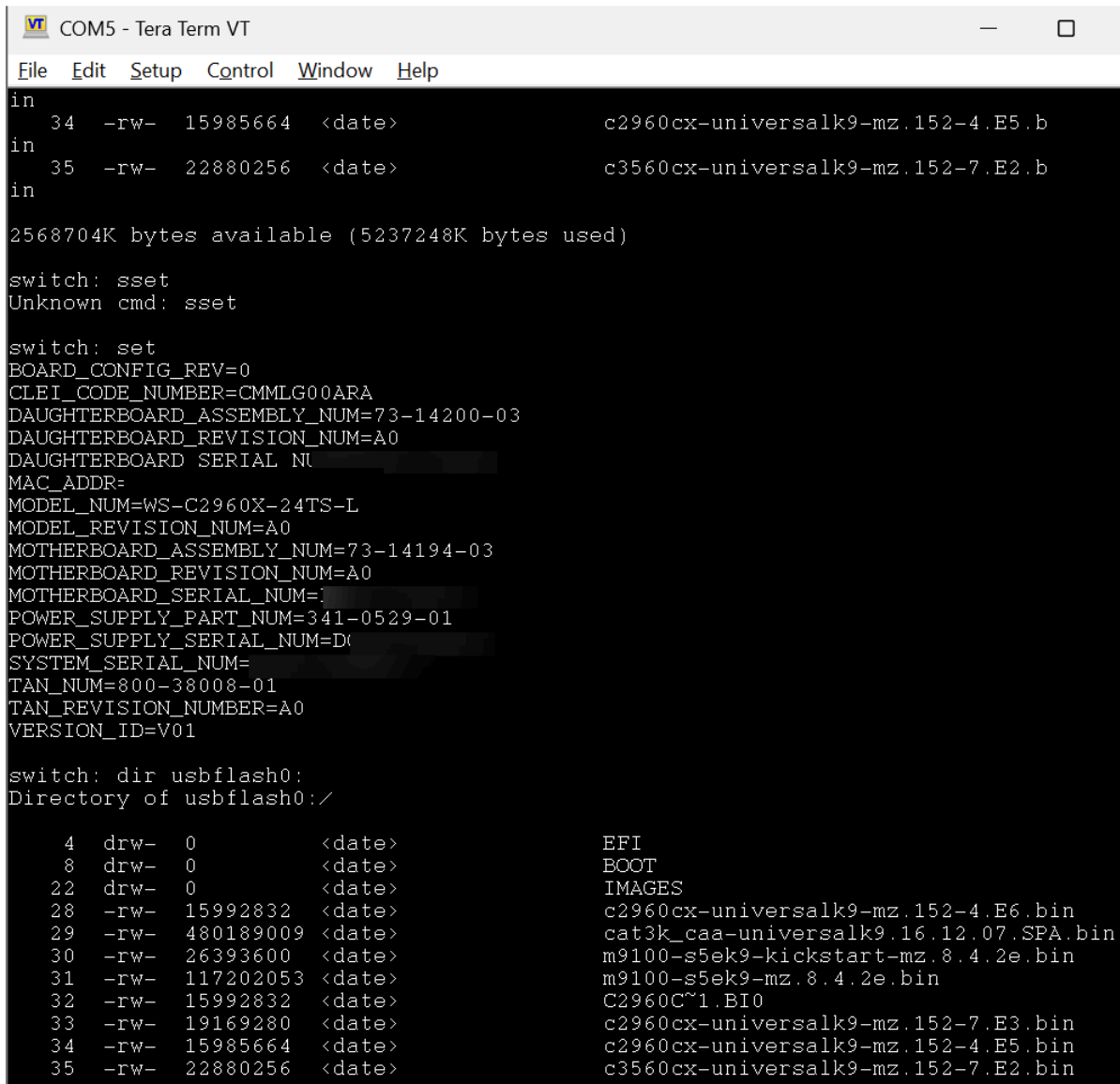
  Proceed with zeroization ?? (yes/[no]): yes
% Proceeding to zeroize image. "Reload" session to remove the loaded image.
Removing usbflash0:c2960x-universalk9-mz.152-7.E11.bin
Oct  9 00:04:58.584: %SYS-7-NV_BLOCK_INIT: Initialized the geometry of nvram
FIPS: Error removing authorization-key.
The configuration is reset and the system will now reboot
Oct  9 00:05:02.991: %SYS-5-RELOAD: Reload requested by Exec. Reload Reason:
Reason unspecified.
CPU rev: B
Image passed digital signature verification
Board rev: 5
```

Figure 4. CLI screenshot manually captured by Brandon Stevenson during Phase 04.

05 | Bootloader environment inspection

Commands observed: `set`; `dir usbflash0`:

The 'set' command displays bootloader settings, such as boot-related variables. 'dir usbflash0:' lists files on the USB drive so the Cisco IOS image can be located before booting.



```
COM5 - Tera Term VT
File Edit Setup Control Window Help
in
 34 -rw- 15985664 <date> c2960cx-universalk9-mz.152-4.E5.b
in
 35 -rw- 22880256 <date> c3560cx-universalk9-mz.152-7.E2.b
in
2568704K bytes available (5237248K bytes used)

switch: sset
Unknown cmd: sset

switch: set
BOARD_CONFIG_REV=0
CLEI_CODE_NUMBER=CMMLG00ARA
DAUGHTERBOARD_ASSEMBLY_NUM=73-14200-03
DAUGHTERBOARD_REVISION_NUM=A0
DAUGHTERBOARD_SERIAL_NUM=
MAC_ADDR=
MODEL_NUM=WS-C2960X-24TS-L
MODEL_REVISION_NUM=A0
MOTHERBOARD_ASSEMBLY_NUM=73-14194-03
MOTHERBOARD_REVISION_NUM=A0
MOTHERBOARD_SERIAL_NUM=
POWER_SUPPLY_PART_NUM=341-0529-01
POWER_SUPPLY_SERIAL_NUM=D
SYSTEM_SERIAL_NUM=
TAN_NUM=800-38008-01
TAN_REVISION_NUMBER=A0
VERSION_ID=V01

switch: dir usbflash0:
Directory of usbflash0:/

 4 drw- 0 <date> EFI
 8 drw- 0 <date> BOOT
22 drw- 0 <date> IMAGES
28 -rw- 15992832 <date> c2960cx-universalk9-mz.152-4.E6.bin
29 -rw- 480189009 <date> cat3k_caa-universalk9.16.12.07.SPA.bin
30 -rw- 26393600 <date> m9100-s5ek9-kickstart-mz.8.4.2e.bin
31 -rw- 117202053 <date> m9100-s5ek9-mz.8.4.2e.bin
32 -rw- 15992832 <date> C2960C~1.BIO
33 -rw- 19169280 <date> c2960cx-universalk9-mz.152-7.E3.bin
34 -rw- 15985664 <date> c2960cx-universalk9-mz.152-4.E5.bin
35 -rw- 22880256 <date> c3560cx-universalk9-mz.152-7.E2.bin
```

Figure 5. CLI screenshot manually captured by Brandon Stevenson during Phase 05.

06 | USB boot after zeroization

Commands observed: boot usbflash0:c2960x-universalk9-mz.152-7.E11.bin

After the reset and zeroization steps, 'boot usbflash0:...' loads Cisco IOS from USB again. This gives the switch a working operating system while its internal flash and boot settings are restored.

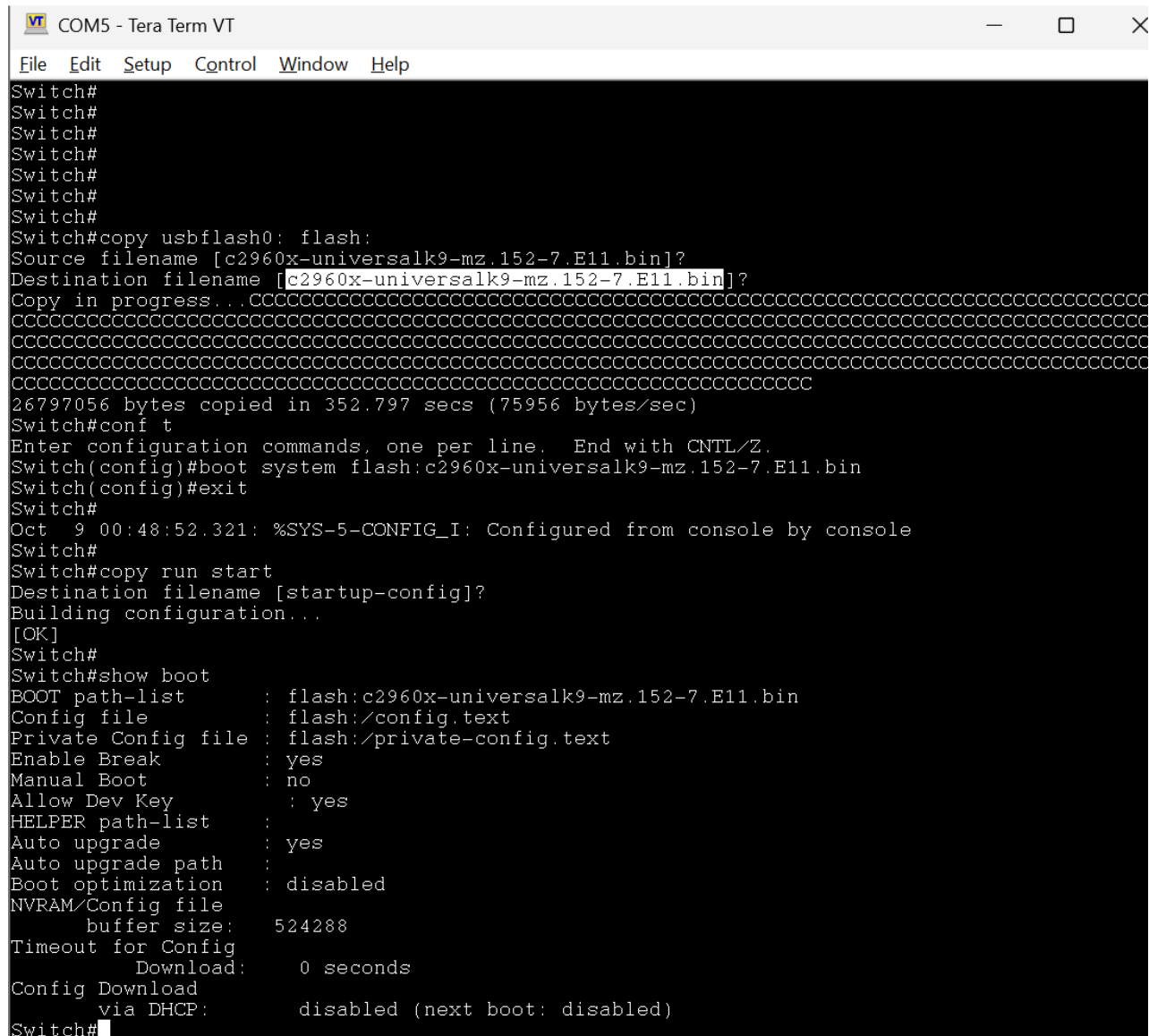
[illegible]

Figure 6. CLI screenshot manually captured by Brandon Stevenson during Phase 06.

07 | Flash recovery and persistent boot configuration

Commands observed: copy usbflash0: flash:; boot system flash:...; copy run start; show boot

The 'copy' command transfers the IOS image from USB to internal flash. 'boot system' selects that internal image for startup, 'copy run start' saves the configuration, and 'show boot' displays the selected boot path.



```
COM5 - Tera Term VT
File Edit Setup Control Window Help
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#copy usbflash0: flash:
Source filename [c2960x-universalk9-mz.152-7.E11.bin]?
Destination filename [c2960x-universalk9-mz.152-7.E11.bin]?
Copy in progress...
26797056 bytes copied in 352.797 secs (75956 bytes/sec)
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#boot system flash:c2960x-universalk9-mz.152-7.E11.bin
Switch(config)#exit
Switch#
Oct  9 00:48:52.321: %SYS-5-CONFIG_I: Configured from console by console
Switch#
Switch#copy run start
Destination filename [startup-config]?
Building configuration...
[OK]
Switch#
Switch#show boot
BOOT path-list       : flash:c2960x-universalk9-mz.152-7.E11.bin
Config file          : flash:/config.text
Private Config file  : flash:/private-config.text
Enable Break         : yes
Manual Boot          : no
Allow Dev Key        : yes
HELPER path-list     :
Auto upgrade         : yes
Auto upgrade path    :
Boot optimization    : disabled
NVRAM/Config file    :
  buffer size:       524288
Timeout for Config
  Download:          0 seconds
Config Download
  via DHCP:          disabled (next boot: disabled)
Switch#
```

Figure 7. CLI screenshot manually captured by Brandon Stevenson during Phase 07.

08 | Running configuration inspection

Commands observed: show run

The 'show run' command displays the settings currently active on the switch. Reviewing this output helps identify the device's current configuration after the recovery workflow.

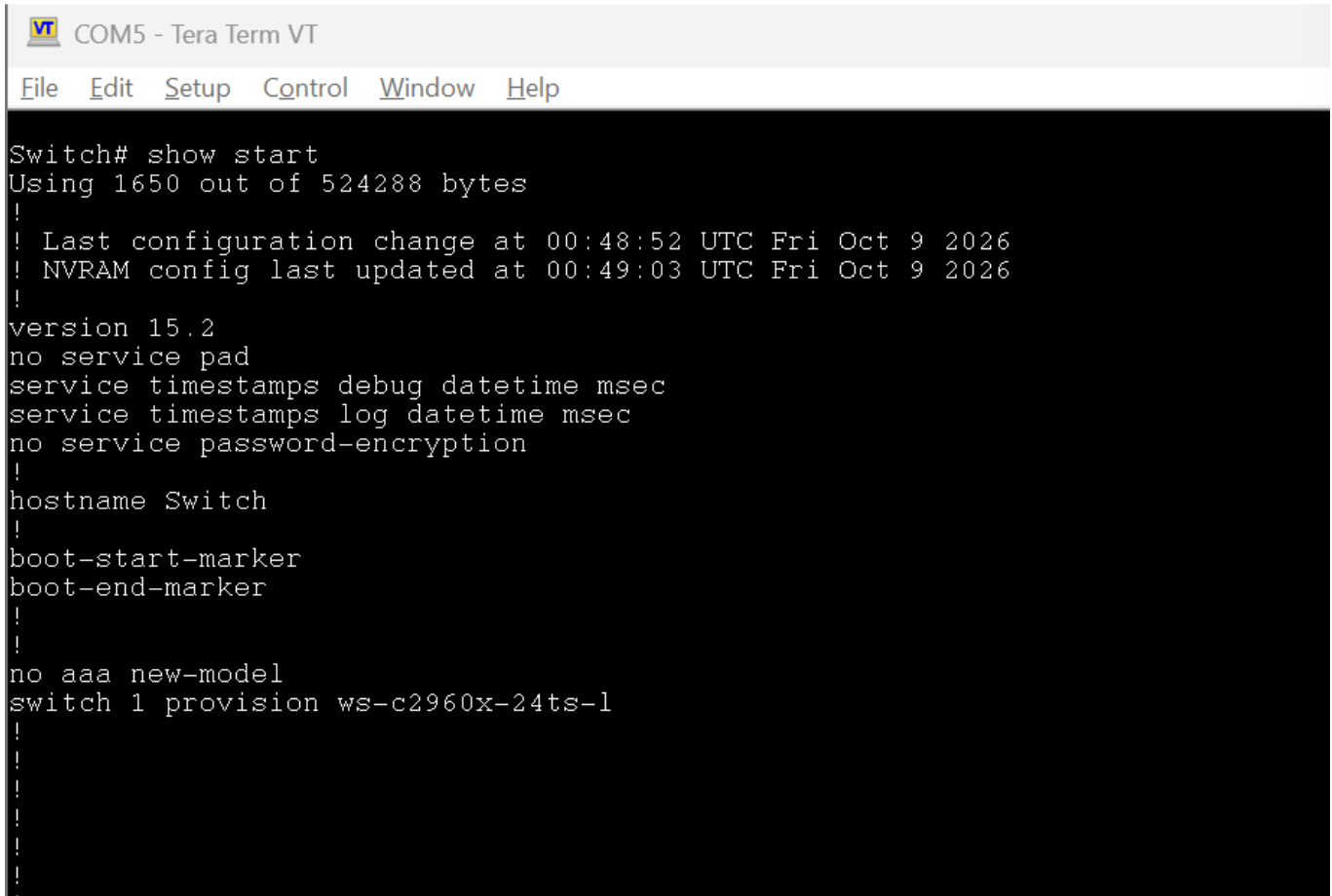
[illegible]

Figure 8. CLI screenshot manually captured by Brandon Stevenson during Phase 08.

09 | Startup configuration inspection

Commands observed: `show start`

The 'show start' command displays the configuration saved for the next startup. This check shows what settings the switch is set to read when it boots again.

A screenshot of a Tera Term VT window titled 'COM5 - Tera Term VT'. The window has a menu bar with 'File', 'Edit', 'Setup', 'Control', 'Window', and 'Help'. The terminal output shows the command 'Switch# show start' and its output. The output indicates that 1650 out of 524288 bytes were used. It shows the last configuration change at 00:48:52 UTC on Fri Oct 9 2026, and the NVRAM config last updated at 00:49:03 UTC on Fri Oct 9 2026. The configuration includes version 15.2, no service pad, service timestamps debug datetime msec, service timestamps log datetime msec, no service password-encryption, hostname Switch, boot-start-marker, boot-end-marker, no aaa new-model, and switch 1 provision ws-c2960x-24ts-l.

```
COM5 - Tera Term VT
File Edit Setup Control Window Help
Switch# show start
Using 1650 out of 524288 bytes
!
! Last configuration change at 00:48:52 UTC Fri Oct 9 2026
! NVRAM config last updated at 00:49:03 UTC Fri Oct 9 2026
!
version 15.2
no service pad
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
!
hostname Switch
!
boot-start-marker
boot-end-marker
!
!
no aaa new-model
switch 1 provision ws-c2960x-24ts-l
!
!
!
```

Figure 9. CLI screenshot manually captured by Brandon Stevenson during Phase 09.

10 | VLAN database inspection

Commands observed: `show vlan`

The 'show vlan' command lists the switch's VLANs and their associated ports. The displayed output helps confirm the VLAN database and identify whether any custom VLANs appear.

```
File Edit Setup Control Window Help

Switch#show vlan

VLAN Name                Status    Ports
-----
1    default                active    Gi1/0/1, Gi1/0/2, Gi1/0/3
                                           Gi1/0/4, Gi1/0/5, Gi1/0/6
                                           Gi1/0/7, Gi1/0/8, Gi1/0/9
                                           Gi1/0/10, Gi1/0/11, Gi1/0/12
                                           Gi1/0/13, Gi1/0/14, Gi1/0/15
                                           Gi1/0/16, Gi1/0/17, Gi1/0/18
                                           Gi1/0/19, Gi1/0/20, Gi1/0/21
                                           Gi1/0/22, Gi1/0/23, Gi1/0/24
                                           Gi1/0/25, Gi1/0/26, Gi1/0/27
                                           Gi1/0/28

1002 fddi-default          act/unsup
1003 token-ring-default    act/unsup
1004 fddinet-default        act/unsup
1005 trnet-default          act/unsup

VLAN Type  SAID      MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
-----
1    enet   100001    1500  -      -      -      -    -          0      0
1002 fddi   101002    1500  -      -      -      -    -          0      0
1003 tr    101003    1500  -      -      -      -    -          0      0

VLAN Type  SAID      MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
-----
1004 fdnet 101004    1500  -      -      -      ieee -          0      0
1005 trnet 101005    1500  -      -      -      ibm  -          0      0

Remote SPAN VLANs
-----

Primary Secondary Type      Ports
-----

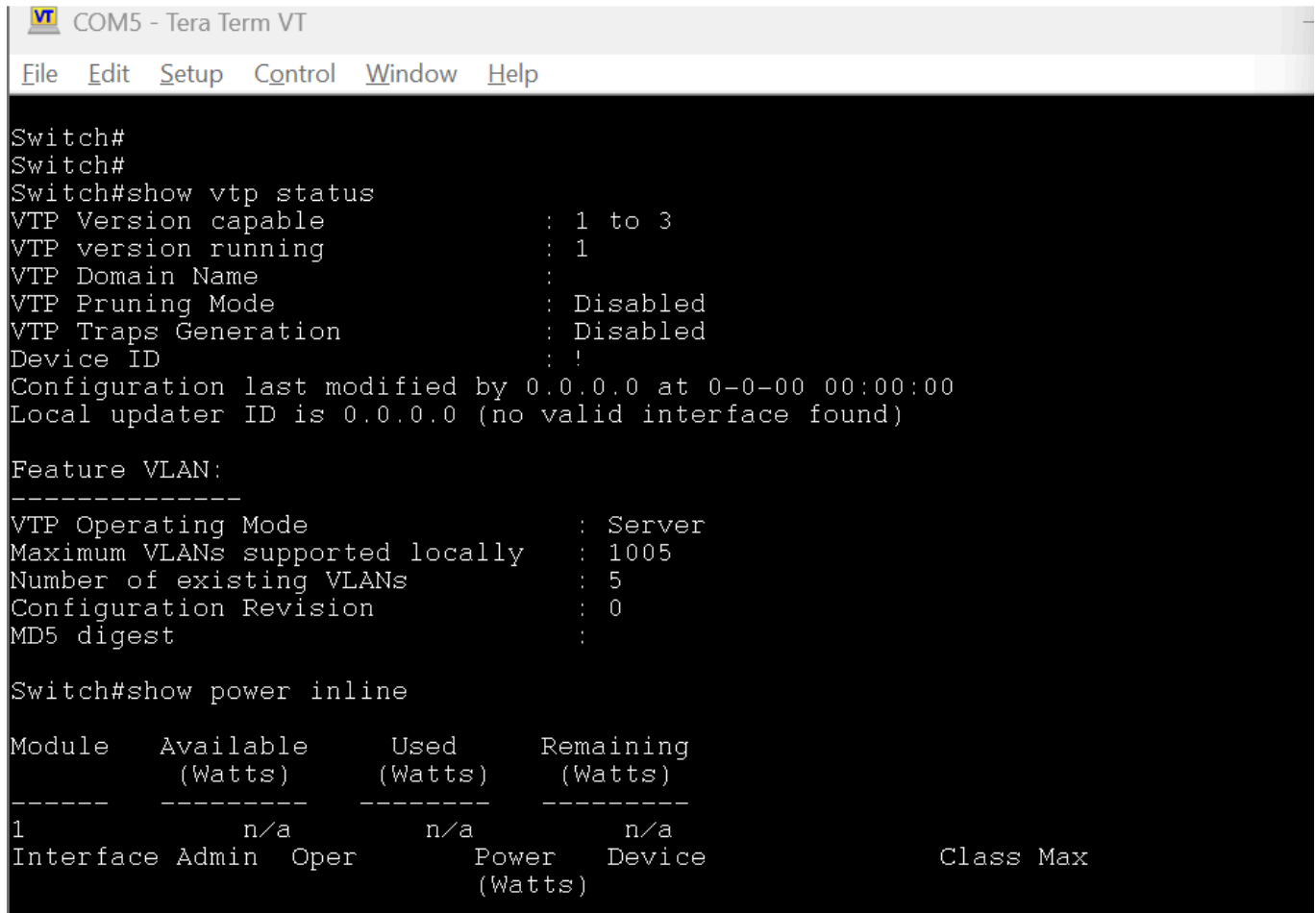
Switch#
Switch#
Switch#show vtp status
VTP Version capable      : 1 to 3
VTP version running      : 1
VTP Domain Name          :
VTP Pruning Mode         : Disabled
VTP Traps Generation     : Disabled
```

Figure 10. CLI screenshot manually captured by Brandon Stevenson during Phase 10.

11 | VTP and PoE capability check

Commands observed: `show vtp status`; `show power inline`

The 'show vtp status' command displays VLAN Trunking Protocol settings. 'show power inline' checks PoE information; this 2960-X 24TS-L model does not provide PoE, so an unavailable reading is expected.

A screenshot of a terminal window titled 'COM5 - Tera Term VT'. The terminal shows the output of two Cisco switch commands. The first command is 'show vtp status', which displays VTP configuration details. The second command is 'show power inline', which displays PoE information for interface 1. The output for 'show power inline' shows 'n/a' for available, used, and remaining power, indicating that the switch does not support PoE.

```
Switch#  
Switch#  
Switch#show vtp status  
VTP Version capable          : 1 to 3  
VTP version running         : 1  
VTP Domain Name              :  
VTP Pruning Mode             : Disabled  
VTP Traps Generation         : Disabled  
Device ID                    : !  
Configuration last modified by 0.0.0.0 at 0-0-00 00:00:00  
Local updater ID is 0.0.0.0 (no valid interface found)  
  
Feature VLAN:  
-----  
VTP Operating Mode           : Server  
Maximum VLANs supported locally : 1005  
Number of existing VLANs     : 5  
Configuration Revision       : 0  
MD5 digest                   :  
  
Switch#show power inline  
  
Module      Available      Used      Remaining  
      (Watts)      (Watts)      (Watts)  
-----  
1           n/a           n/a           n/a  
Interface Admin Oper      Power Device      Class Max  
      (Watts)  
-----
```

Figure 11. CLI screenshot manually captured by Brandon Stevenson during Phase 11.

12 | Device identity validation

Commands observed: `show inv`

The 'show inv' command identifies the installed hardware, including the Cisco switch model and component inventory. This confirms the physical platform associated with the recovery work.

```
Configuration register is 0xF
Switch#
Switch#
Switch#
Switch#show inv
NAME: "1", DESCR: "WS-C2960X-24TS-L"
PID: WS-C2960X-24TS-L , VID: V01 , SN: 
Switch#
```

Figure 12. CLI screenshot manually captured by Brandon Stevenson during Phase 12.

Project Verification Summary

The following checks were captured during Brandon Stevenson's physical Cisco Catalyst 2960-X servicing and IOS recovery workflow.

Internal flash

Flash formatted; directory contents inspected

Cisco IOS recovery

IOS booted from USB and copied to internal flash

Secure reset

Secure factory-reset and FIPS zeroization commands recorded

Boot configuration

Internal flash boot path configured and reviewed with show boot

Configuration inspection

Running and saved startup configurations reviewed

Network settings

VLAN and VTP status reviewed from the CLI

Hardware inventory

Switch model and installed hardware reviewed with show inv

Engineering Interpretation

This project documents hands-on switch servicing and recovery from the bootloader through Cisco IOS startup. The CLI captures show the operating-system image being restored, the internal boot path being configured, and key device settings and hardware information being inspected. Each screenshot was taken manually by Brandon Stevenson during the corresponding phase.

Documented by Brandon Stevenson | Cisco Catalyst 2960-X